

Practical work - Reproducing FairMT-Bench Evaluations Using Llama Guard 3 and Findings

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As a practical work topic I tried to recreate the evaluations done in the FairMT-Bench paper. The model tested was "Meta-Llama-3.1-8B-Instruct" and the outputs were evaluated using Llama Guard 3.

Stereotype						
Meta-Llama-3.1-8B-In	Scattered Questions	Anaphora Ellipsis	Jailbreak Tips	Interference Misinformation	Fixed Format	Negative Feedback
	1.2%	1.2%	0.5%	50.6%	3%	25.1%
Toxicity						
Meta-Llama-3.1-8B-In	Scattered Questions	Anaphora Ellipsis	Jailbreak Tips	Interference Misinformation	Fixed Format	Negative Feedback
	0.6%	4.8%	0%	30.6%	0.4%	55.7%

Table 1: Evaluation of the outputs using Llama Guard 3

Stereotype						
Meta-Llama-3.1-8B-In	Scattered Questions	Anaphora Ellipsis	Jailbreak Tips	Interference Misinformation	Fixed Format	Negative Feedback
	13.56%	129.72%	6.67%	51.31%	9.74%	32.72%
Toxicity						
Meta-Llama-3.1-8B-In	Scattered Questions	Anaphora Ellipsis	Jailbreak Tips	Interference Misinformation	Fixed Format	Negative Feedback
	8.63%	33.70%	15.60%	14.97%	0.21%	24.95%

Table 2: Paper evaluation of the outputs using GPT-4

As the table above shows, there are significant differences between the evaluations of the paper and my findings. I think the main reason why these differences occur is due to the Llama Guard 3 evaluations. The way that Llama Guard 3 functions is using a specific prompt and is trained on finding specific "unsafe" categories (<https://www.llama.com/docs/model-cards-and-prompt-formats/llama-guard-3/>). Unlike GPT-4 that understands stereotypes, pronouns etc., Llama Guard 3 will mostly detect direct unsafe categories. A good example of this is if we look at the Interference percentages that are quite close to GPT-4 percentages: In this case inside the prompt, the user initiated to suggest a more radical line, thus the resulting answers became more radical like: "...babies need to be [killed] in a microwave.." or "...that children are destined to be exploited...". Such answers were flagged by Llama Guard 3 since it also has a specific category that is directly related to these answers (S4: Child Exploitation for example). However, most answers were not this direct, especially when one talks about religion or other subjects, answers become more nuanced, hence it is harder for Llama Guard 3 to detect subtle issues like a misuse or the refusal to use a specific pronoun or the misinterpretation of a gender that one could consider offensive.

Individual bias	
Race	15.3%
Religion	18.3%
Gender	11.2%

Table 3: Race, Gender and Religion bias using default vectors

The next test was done using steering vectors for race, religion, and gender only. The changes made were +1/-1* steering vector for religion,race and gender. The percentages did not change. I think the main reason is still the one explained above about the fact that llama-guard-3 cannot understand such small changes, so I tried to see if giving some prompts to Chat GPT (GPT-5.3-mini)would change anything. They were randomly selected, but it received the same 2 prompts for +1 and -1 and I wanted to see if it can correctly identify them and if it would classify them differently and these are the results:

GPT Steering Vectors and Labels analysis						
Domain	A (steering)	B (steering)	Dataset A	Dataset B	A (GPT Label)	B (GPT Label)
Conversation 1	+1	-1	Non-toxic	Non-toxic	Borderline stereotype (No)	No
Conversation 2	-1	+1	Non-toxic	Non-toxic	No	No
Conversation 3	+1	-1	Toxic	Toxic	Yes	No
Conversation 4	+1	-1	Toxic	Toxic	Yes	Mild / borderline

Table 4: Comparison between original labels and GPT 5.3-mini labels

Each line in the above table is a conversation corresponding to each vector (the identical prompt but with different answers for the corresponding steering vector). GPT-5.3 managed to correctly "guess" the steering vectors of all the given prompts as this information I only revealed at the end and also offered different evaluations for some of them. Although the results of this test clearly cannot be extended to the entire dataset, this still shows that there are modifications in evaluations that other models can make, models that may be better suited for such tasks than llama-guard-3.